08 — Mu-Calculus

Fixpoint Operators

Operator	Semantics
`nu X. phi(X)`	Greatest fixpoint — starts from *all states* and removes violators downward. Captures **safety** and **invariance**.
`mu X. phi(X)`	Least fixpoint — starts from *no states* and adds satisfiers upward. Captures **reachability** and **liveness**.

Intuition: `nu` keeps states that *always* satisfy φ; `mu` finds states that *eventually* satisfy φ.
Fixpoint variables (`X`) are recursive: the formula "unfolds" until a stable set is reached.

Modalities

Syntax	Meaning
`[] phi`	Box: *all* successors satisfy φ
`<> phi`	Diamond: *some* successor satisfies φ

Without labels, modalities quantify over **all** outgoing transitions from the current state.

Boolean Operators

`phi && psi`	Conjunction (and)
`phi || psi`	Disjunction (or)
`!  phi`	Negation (not)
`true`	Holds in every state
`false`	Holds in no state

Labeled Modalities

Restrict modalities to specific labels:

```
// All a-or-b transitions lead to phi
[ labels = {a, b} ] phi

// Some a-transition leads to phi
< labels = {a} > phi
```

- Without `labels = {...}`: considers **all** transitions.
- With labels: considers **only** transitions on the specified labels.
- Multiple labels in the set act as a union.

Nesting and Alternation

Formulas can nest fixpoints and modalities:

```
// Alternation: nu outside, mu inside
nu X. (mu Y. (p || <> Y)) && ([] X)
```

- **Alternation depth** affects complexity — deeper nesting is more expensive.
- Inner `mu` resets each time the outer `nu` iterates.
- Most practical properties have alternation depth ≤ 2.

Common Patterns

Pattern	Formula
Safety invariant	`nu X. ([] X)`
Reachability	`mu X. (target || <> X)`
GF (inf. often)	`nu X. (mu Y. (p || <> Y)) && ([] X)`
No deadlock	`nu X. (<> true) && ([] X)`
Labeled check	`(!  src) || [ labels = {a} ] dst`
Mutual exclusion	`(!  A) || (!  B)`
Recovery	`(!  bad) || (mu Y. (good || <> Y))`
Ctrl reachability	`mu X. (p || [ (ctrl = controllable) ] X)`

Reading Patterns

- **Safety** = "always stays in good states" $\Rightarrow$ `nu` + `[]`
- **Reachability** = "eventually reaches target" $\Rightarrow$ `mu` + `<>`
- **Liveness** = "always eventually" $\Rightarrow$ `nu` wrapping `mu`
- Implications (`p` $\Rightarrow$ `q`) encoded as `(!  p) || q`

Common Mistakes

- Swapping `nu`/`mu` — reverses safety vs. reachability semantics.
- Using `<>` (diamond) when `[]` (box) is needed — "some path" vs. "all paths".
- Forgetting that `[] false` holds vacuously in deadlock states (no successors).